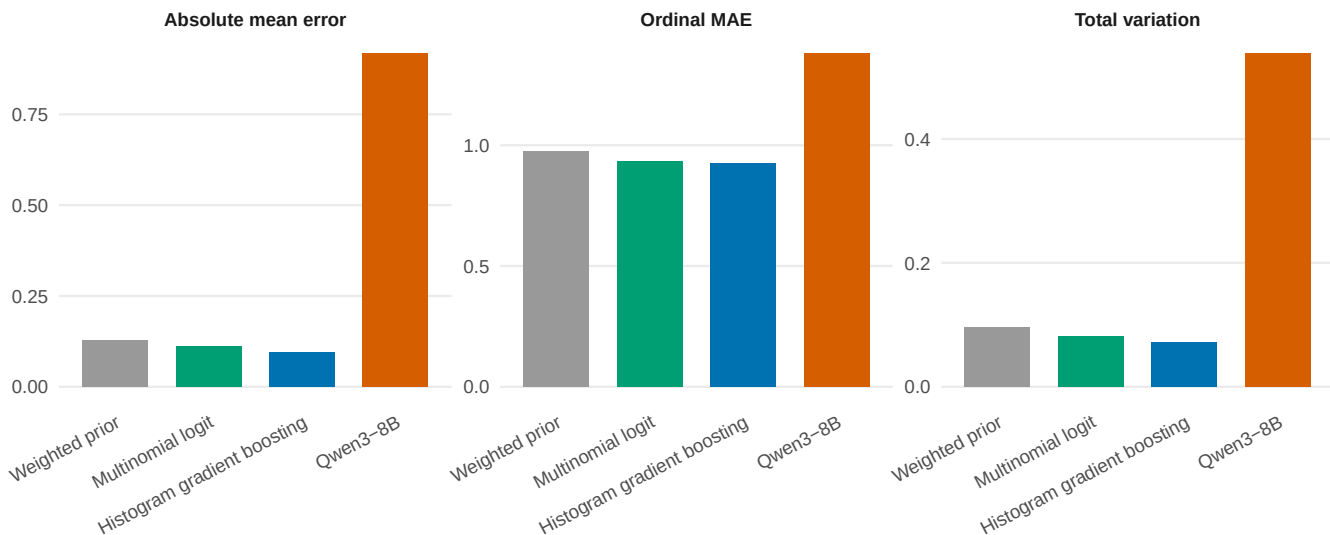


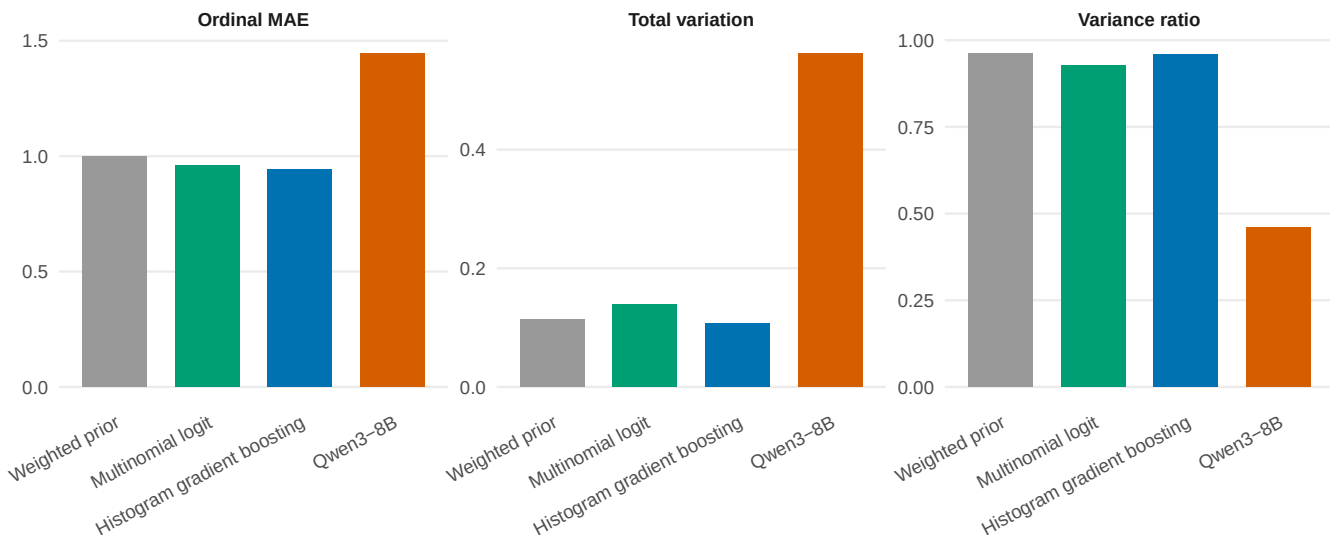
Outcome-informed benchmarks separate sparse-profile limits from generator distortion

A. Same 300 profiles: outcome-informed references recover the population



B. Temporal transfer: train on 2012–2018, evaluate 2021

Qwen3-8B is zero-shot; reference models use historical outcomes only.



Notes: Lower is better for MAE, absolute mean error, and total variation; a variance ratio of 1 indicates recovery. Reference models observe survey outcomes during training, whereas Qwen3-8B does not.